Answer Key - Health Sciences - Graduate Level Assessment 12 (Answers are ordered exactly as in the question paper.)

Multiple Choice

1. **Answer:** C
   Options: A) fasting blood glucose $\geq$ 160 mg/dl; B) fasting blood glucose ? 140 mg/dl; C) fasting blood glucose $\geq$ 126 mg/dl; D) random glucose > 180 mg/dl; E) fasting blood glucose ? 120 mg/dl
   Note: The fasting blood glucose level of 126 mg/dl or higher is established by the American Diabetes Association as the diagnostic threshold for diabetes, indicating impaired glucose regulation.

2. **Answer:** C
   Options: A) studying the effects of radiation exposure; B) identifying the presence of bacterial infections; C) solving criminal and paternity cases; D) determining blood type; E) transferring disease resistance factors into bone marrow cells
   Note: Simple tandem repeat polymorphisms are highly variable regions of DNA that can be used to create unique genetic profiles for individuals, making them particularly useful in forensic science for solving criminal cases and establishing biological relationships in paternity testing.

3. **Answer:** A
   Options: A) synovial fluid and articular discs.; B) capsule and ligaments.; C) articular cartilage and ligaments.; D) capsule and articular cartilage.; E) synovial membrane and capsule.
   Note: Proprioceptive nerve endings in synovial joints are primarily found in the synovial fluid and articular discs, as these structures are crucial for sensing joint position and movement due to their rich innervation and role in proprioception.

4. **Answer:** A
   Options: A) Weight-for-age Z score <-2 and oedema; B) Height-for-age Z score <-2 or weight-for-height Z score <-2 and oedema; C) Height-for-age Z score <-3 and oedema; D) Weight-for-age Z score <-2 or height-for-age Z score <-2 or oedema; E) Height-for-age Z score <-3 or weight-for-age Z score <-3 and oedema
   Note: Severe acute malnutrition in young children is characterized by a weight-for-age Z score of less than -2, indicating significant underweight, coupled with the presence of oedema, which reflects severe nutritional deficiency and indicates a critical health condition requiring immediate intervention.

5. **Answer:** A
   Options: A) Female catheters are used more frequently than male catheters.; B) Male catheters are bigger than female catheters.; C) Male catheters are more flexible than female catheters.; D) Male catheters are made from a different material than female catheters.; E) Female catheters are longer than male catheters.
   Note: The correct answer highlights the higher prevalence of urinary conditions in

females, which leads to a greater frequency of catheterization procedures for women compared to men.

True / False

6. **Answer:** False
   Options: A) True; B) False
   Note: The Grade 2/5 power indicates that a muscle can move a limb through the full range of motion without resistance, which includes movement against gravity but also in the presence of support or assistance, making the statement false.

7. **Answer:** False
   Options: A) True; B) False
   Note: Research indicates that a higher level of education is associated with a lower risk of Alzheimer's disease in older adults, as education may enhance cognitive reserve and resilience against neurodegenerative conditions.

8. **Answer:** True
   Options: A) True; B) False
   Note: The temporal bone contains the zygomatic process, which extends to articulate with the zygomatic bone, thereby forming the zygomatic arch.

9. **Answer:** True
   Options: A) True; B) False
   Note: Type I muscle fibers, also known as slow-twitch fibers, are rich in myoglobin and mitochondria, enabling them to generate energy through aerobic metabolism, which supports prolonged, endurance-based activities.

10. **Answer:** False
    Options: A) True; B) False
    Note: The concentration of a substance in food alone does not determine safety, as the toxicity of a substance also depends on factors like individual susceptibility, exposure duration, and the presence of other compounds.

Long Form Response

11. **Answer:** DiGeorge/Shprintzen syndrome is primarily caused by a microdeletion on chromosome 22 q 11.2, which affects the development of multiple organ systems, particularly the heart, thymus, and parathyroid glands. This genetic alteration leads to a range of health issues, including congenital heart defects, immune deficiencies due to thymic hypoplasia, and hypoparathyroidism, resulting in calcium imbalances. The syndrome exemplifies how a single chromosomal deletion can produce a spectrum of developmental abnormalities, highlighting the importance of genetic factors in health outcomes. Understanding the genetic basis of this condition is crucial for early diagnosis, management, and genetic counseling for affected families.

Note: correct because it accurately identifies the 22 q 11.2 microdeletion as the genetic cause of DiGeorge/Shprintzen syndrome and details its significant impact on the development and function of critical organ systems, leading to various health complications.

12. **Answer:** Ubiquitin plays a crucial role in the regulation of proteolysis, primarily through its attachment to proteins designated for synthesis rather than catabolism. This small protein tag marks substrates for degradation via the ubiquitin-proteasome pathway, which is essential for maintaining cellular homeostasis by removing misfolded or damaged proteins. In contrast, proteins earmarked for synthesis are not ubiquitinated, allowing for their accumulation and proper function within cellular processes. This selective tagging mechanism underscores the importance of ubiquitin in modulating protein turnover and ensuring that only those proteins that are no longer needed or malfunctioning are degraded, thereby maintaining efficient cellular regulation and function. Ultimately, the balance between ubiquitination and deubiquitination is critical for cellular health, influencing various pathways including stress responses, cell cycle progression, and apoptosis.
    Note: The answer correctly emphasizes that ubiquitin tags proteins for degradation, which is essential for cellular regulation and homeostasis, while proteins destined for synthesis remain non-ubiquitinated to ensure their proper function and accumulation.

End of Answer Key